

2 (**short**) The test frame shown in Fig. 2a experiences loads H and V as shown, applied at the pin-joint P . The frame comprises two identical struts, of Young's Modulus E , cross-sectional area A , second moment of area I and length L . The struts are rigidly clamped at the ends opposite the pin joint, and are perpendicular.

(a) Figure 2b shows the horizontal strut on its own. Write down the relationships between forces T and S , applied in parallel and perpendicular directions at the tip of the strut as shown, and the tip deflections t and s in the same directions. [4]

(b) By rearranging your answers to part (a) to predict the forces required to give certain tip deflections, derive an expression for the displacements h and v arising from application of the forces H and V to the structure in Fig. 2a. [6]

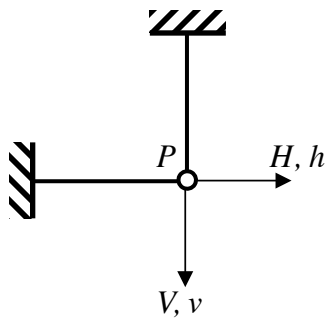


Fig. 2a

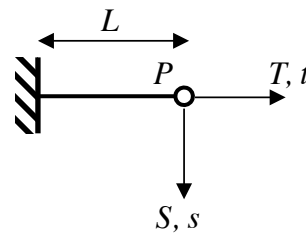


Fig. 2b